COMPLEX CONJUGATE

Conjugate vs negation

Confusing the two operations leads to errors, particularly when simplifying.

Conjugate

RULE

$\overline{a+bi} = a - bi$

WHAT CHANGES

Only the sign of the imaginary part; the real part is kept exactly

EXAMPLE

$\overline{3+2i} = 3 - 2i$

WRITTEN

$\bar{z}$ in pure mathematics and most textbooks; z^* in physics and engineering

Negation

RULE

$-(a+bi) = -a - bi$

WHAT CHANGES

Both signs: the real part and the imaginary part

EXAMPLE

$-(3+2i) = -3 - 2i$

WRITTEN

$-z$

≠ NOT THE SAME

The conjugate is not the same as negation: $3 - 2i$ and $-3 - 2i$ are different numbers.

A real number is its own conjugate, $\overline{5} = 5$, but not its own negative.